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Banister et al.

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(54) **SISO DECODER**

7,093,179 B2 * 8/2006 Shea 714/755

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H03M 13/00 (2006.01)

(52) **U.S. Cl.** **714/780**

(58) **Field of Classification Search** **714/780**
See application file for complete search history.

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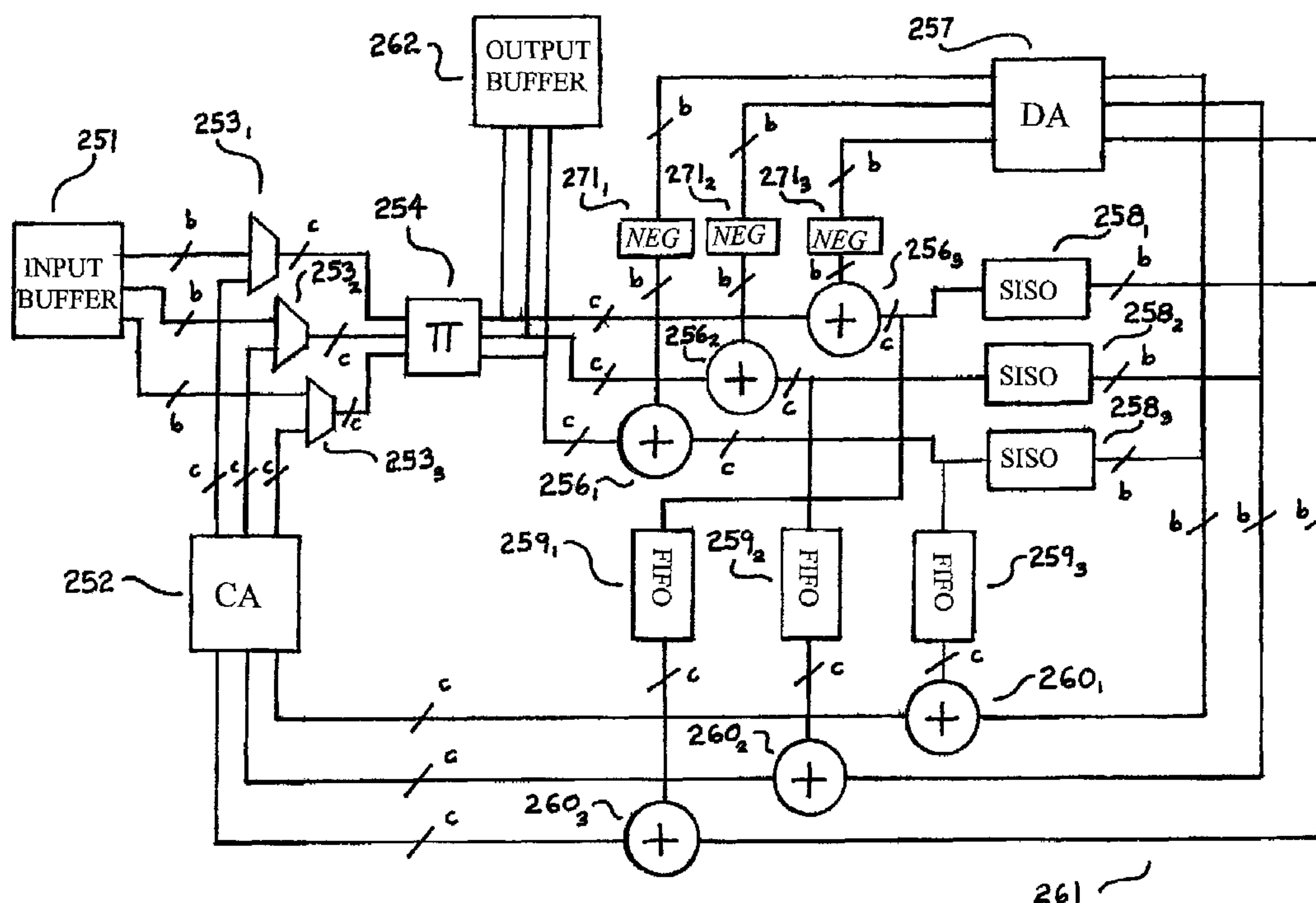
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(57) **ABSTRACT**

The current invention involves a forward error detection system, especially for use with Low Density Parity Check codes. A parallel SISO structure allows the decoder to process multiple parity equations at the same time. There is a new SISO decoder which allows for the updating of the Log-likelihood-ratio's in a single operation, as opposed to the two pass traditionally associated with the Tanner Graph's. In the decoder, there is a mapping structure that correctly aligns the stored estimates, the stored differences and the SISO's. There is also the ability to deal with multiple instances of the same data being processed at the same time. This structure manages the updates and the differences in such a manner that all calculations on a single piece of data that are processed in parallel are incorporated correctly in the new updated estimates.

12 Claims, 13 Drawing Sheets



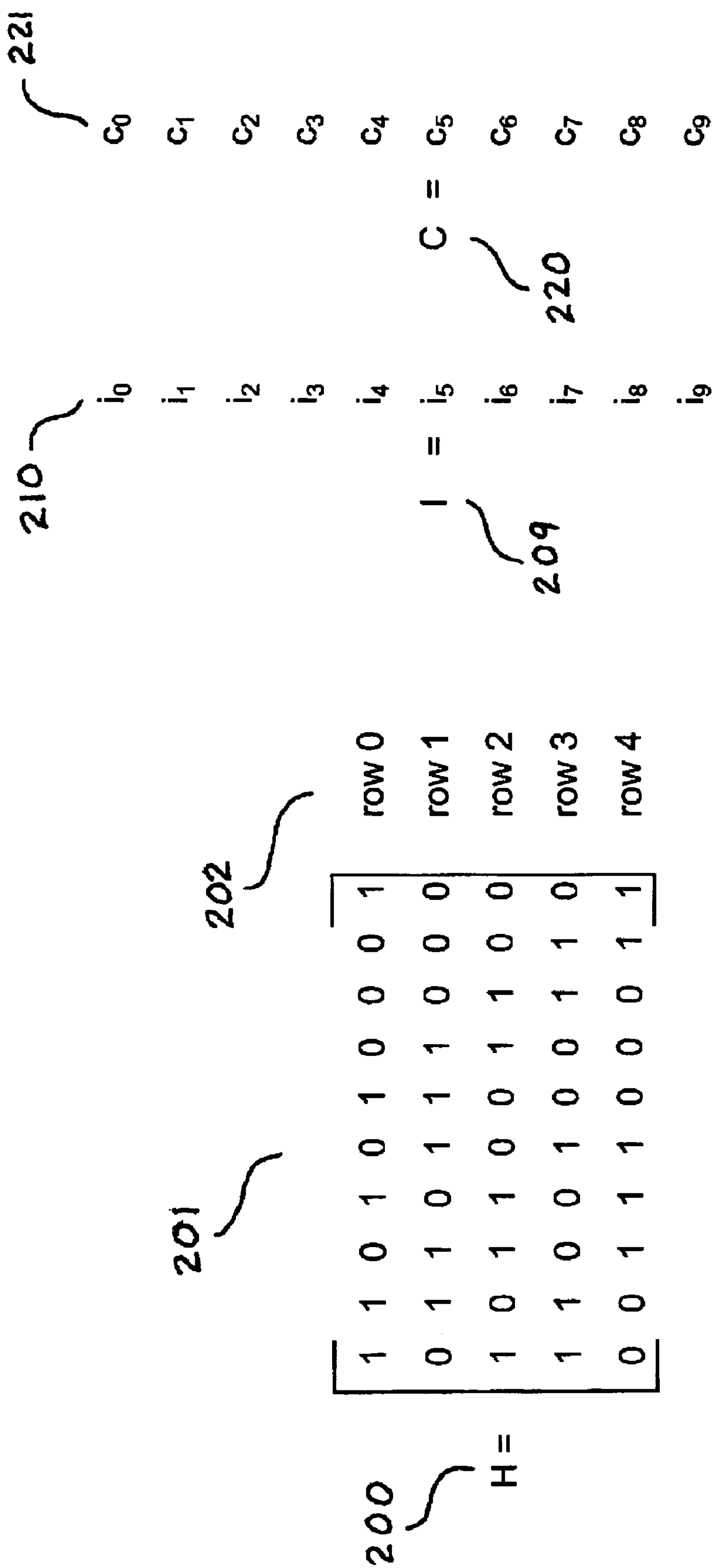


Figure 1

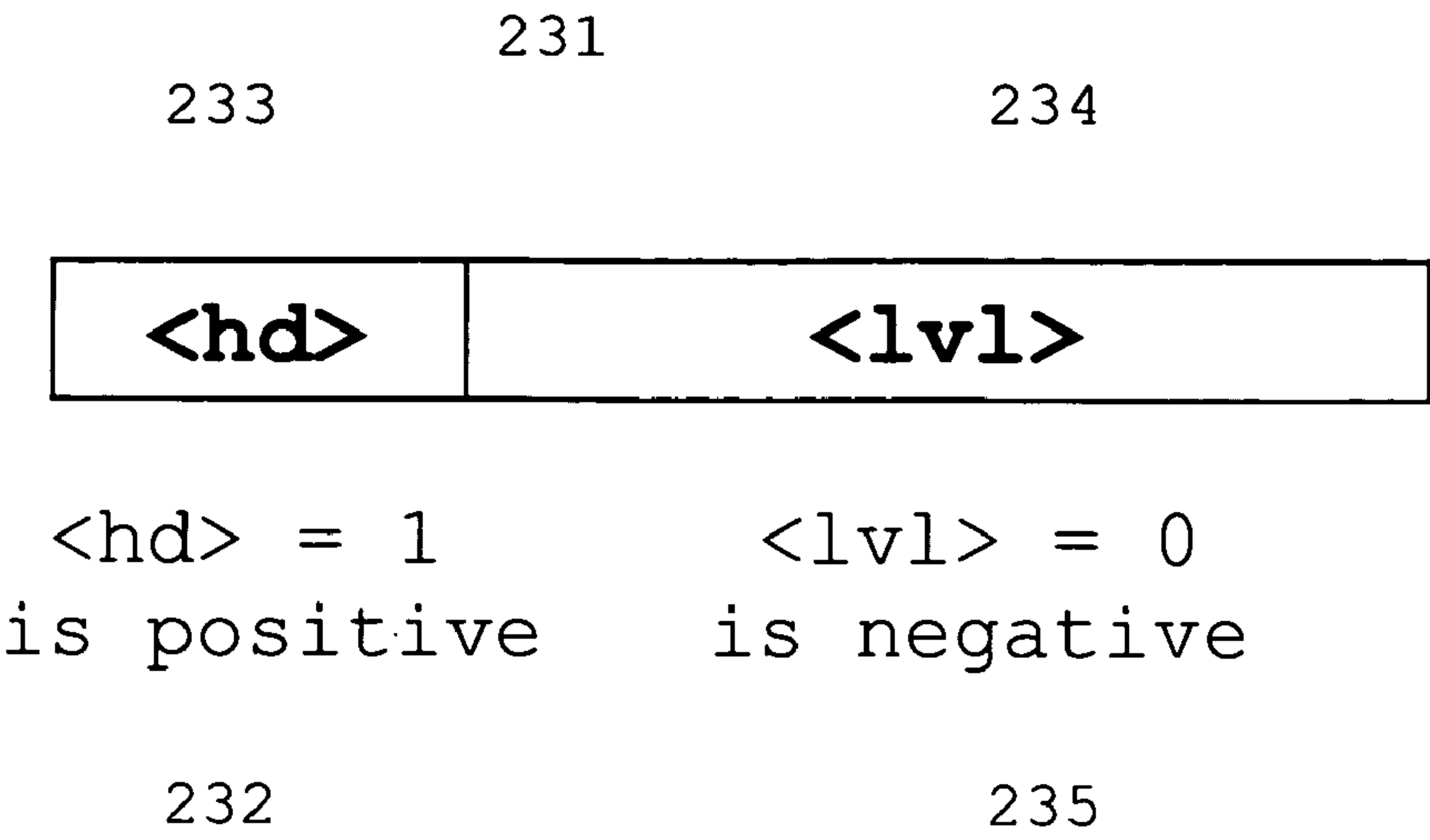


Fig. 2

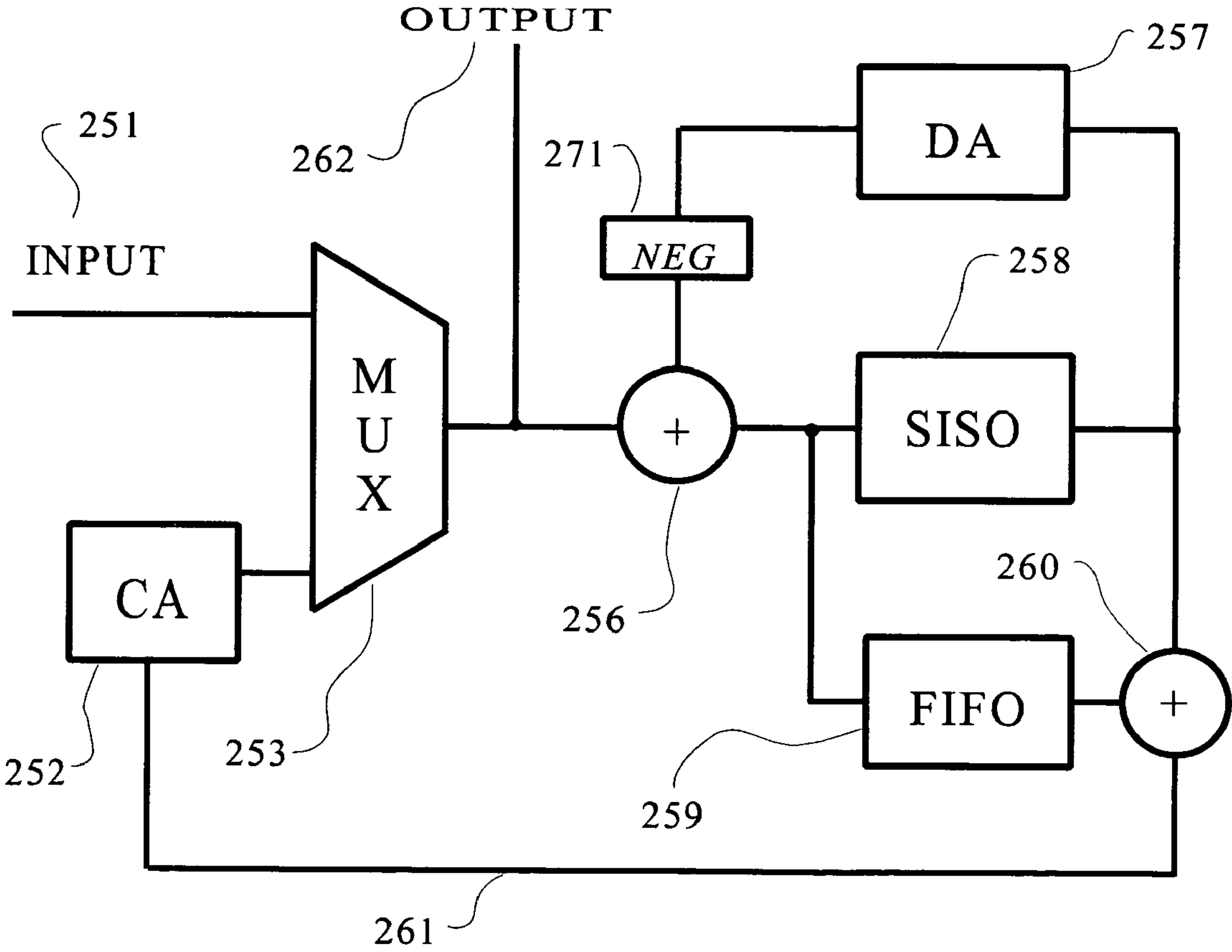


Fig. 3a

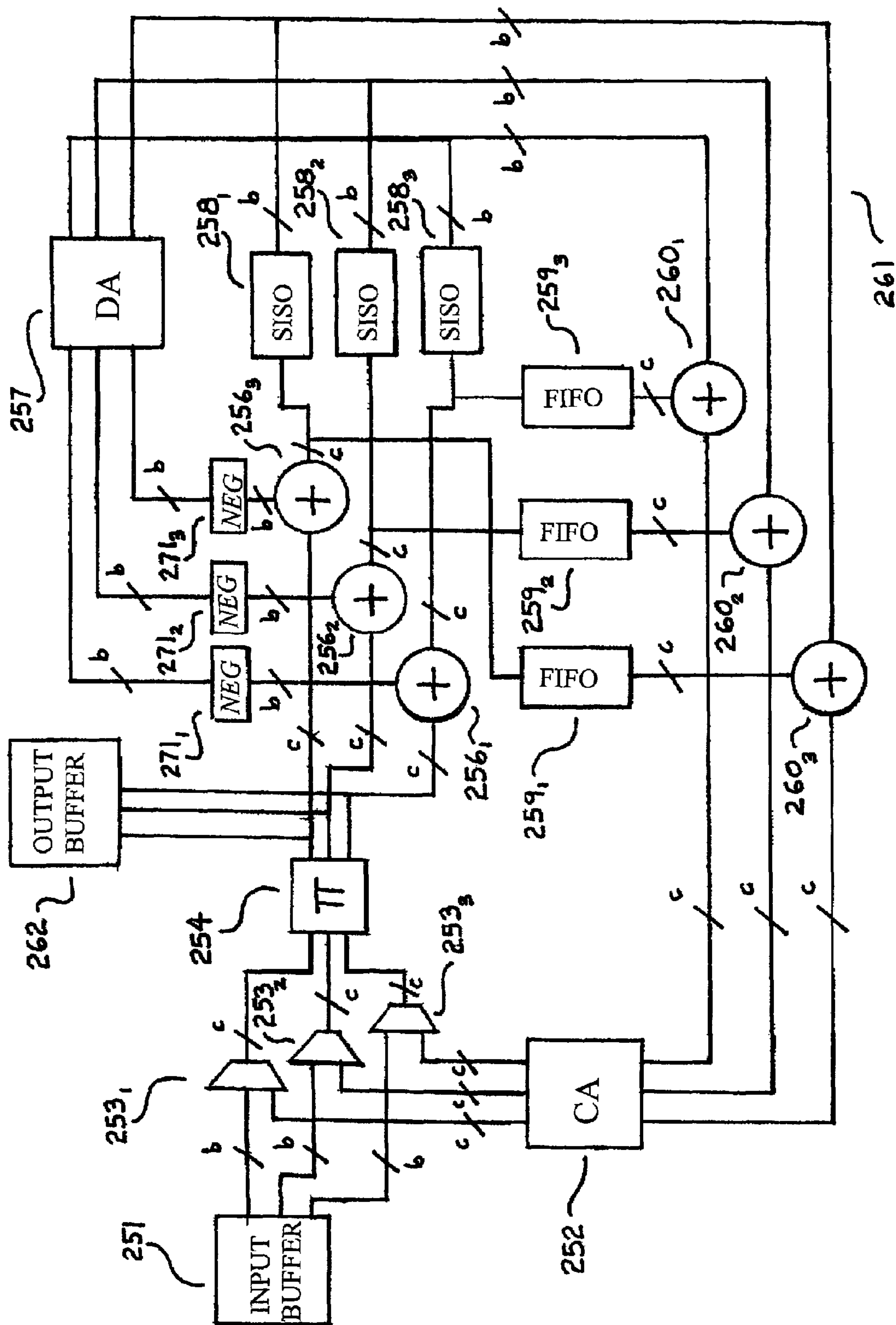
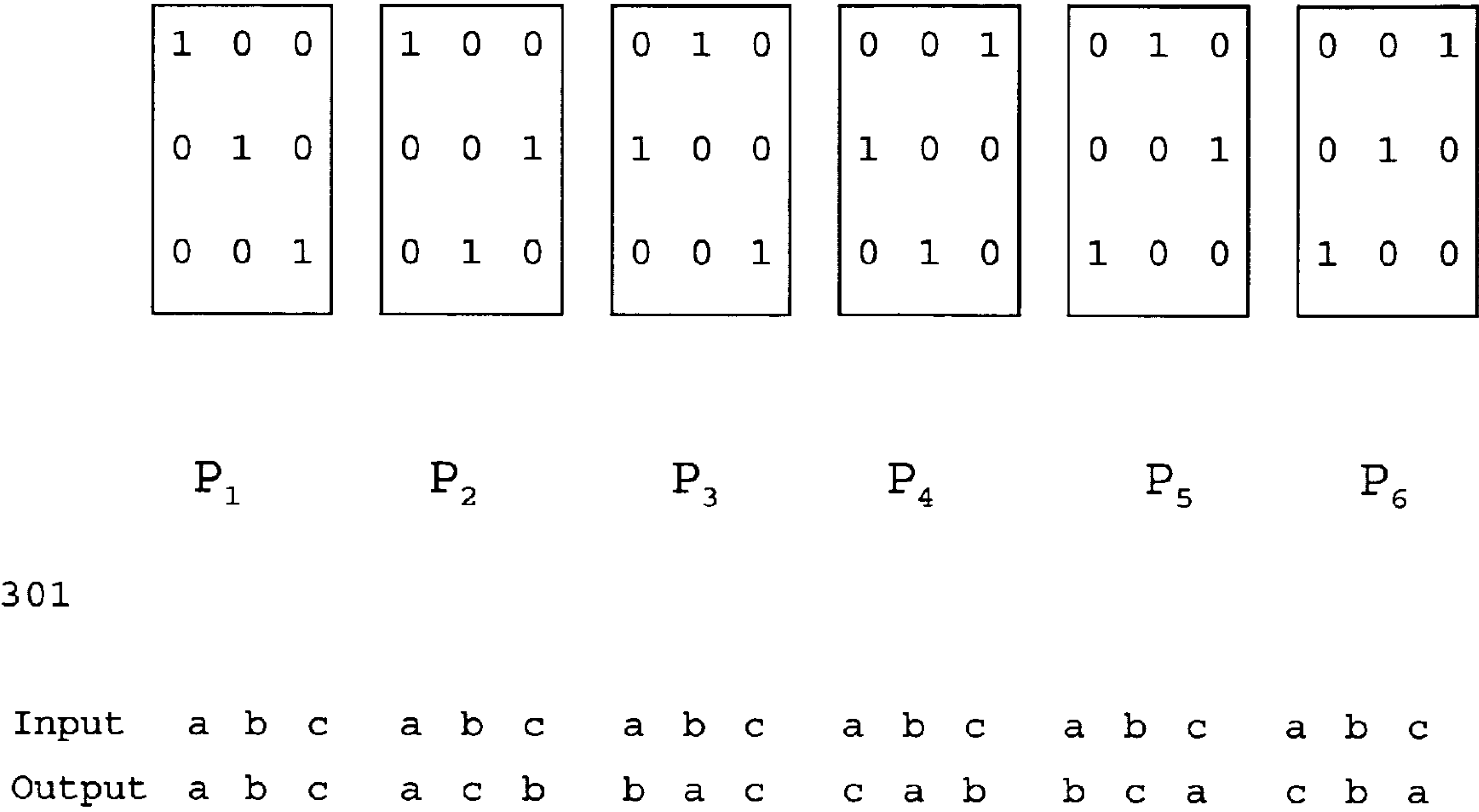


Fig. 3b



302

Fig. 4
Possible permutations for 3 items

[illegible][illegible]

342		332					331				
	ROW ₀	2	3	0	5	0	1	0	0	0	4
	ROW ₁	0	6	4	0	3	1	1	0	0	0
	ROW ₂	4	0	3	6	0	0	1	1	0	0
	ROW ₃	5	2	0	0	6	0	0	1	1	0
	ROW ₄	0	0	5	2	4	0	0	0	1	1

Fig. 6

342

$$\begin{aligned} \text{ROW}_0 &= (0,2) + (1,3) + (3,5) + (5,1) + (9,4) \\ \text{ROW}_1 &= (1,6) + (2,4) + (4,3) + (5,1) + (6,1) \\ \text{ROW}_2 &= (0,4) + (2,3) + (3,6) + (6,1) + (7,1) \\ \text{ROW}_3 &= (0,5) + (1,2) + (4,6) + (7,1) + (8,1) \\ \text{ROW}_4 &= (2,5) + (3,2) + (4,4) + (8,1) + (9,1) \end{aligned}$$

343

Fig. 7

364				361		363
			A			
			1 2 3 4 5 6			
	1	1	2	3	4	5 6
	2	2	1	5	6	3 4
B	3	3	4	1	2	6 5
	4	4	3	6	5	1 2
	5	5	6	2	1	4 3
	6	6	5	4	3	2 1
362						360

Fig. 8

Permutation A followed by Permutation B

373

371

372

	ROW 0	ROW 1	ROW 2	ROW 3	ROW 4
C ₀	P ₂	P ₂	P ₄	P ₅	P ₅
C ₁	P ₃	P ₆	P ₆	P ₂	P ₂
C ₂	P ₁	P ₄	P ₃	P ₃	P ₅
C ₃	P ₅	P ₅	P ₆	P ₆	P ₂
C ₄	P ₁	P ₃	P ₃	P ₆	P ₄
C ₅	P ₁	P ₁	P ₁	P ₁	P ₁
C ₆	P ₁	P ₁	P ₁	P ₁	P ₁
C ₇	P ₁	P ₁	P ₁	P ₁	P ₁
C ₈	P ₁	P ₁	P ₁	P ₁	P ₁
C ₉	P ₄	P ₄	P ₄	P ₄	P ₁

Fig. 9

**Permutations for a complete iteration as
they are stored in CA memory after
each ROW equation is evaluated**

382	$D_{0,0}$	P_2	383
	$D_{0,1}$	P_3	
	$D_{0,3}$	P_5	
	$D_{0,5}$	P_1	
	$D_{0,9}$	P_4	
	$D_{1,1}$	P_6	
	$D_{1,2}$	P_4	
	$D_{1,4}$	P_3	
	$D_{1,5}$	P_1	
	$D_{1,6}$	P_1	
381	$D_{2,0}$	P_4	
	$D_{2,2}$	P_3	
	$D_{2,3}$	P_6	
	$D_{2,6}$	P_1	
	$D_{2,7}$	P_1	
	$D_{3,0}$	P_4	
	$D_{3,4}$	P_3	
	$D_{3,5}$	P_6	
	$D_{3,7}$	P_1	
	$D_{3,8}$	P_1	
	$D_{4,2}$	P_5	
	$D_{4,3}$	P_2	
	$D_{4,4}$	P_4	
	$D_{4,8}$	P_1	
	$D_{4,9}$	P_1	

Fig. 10

Permutations as they are stored in the DA memory

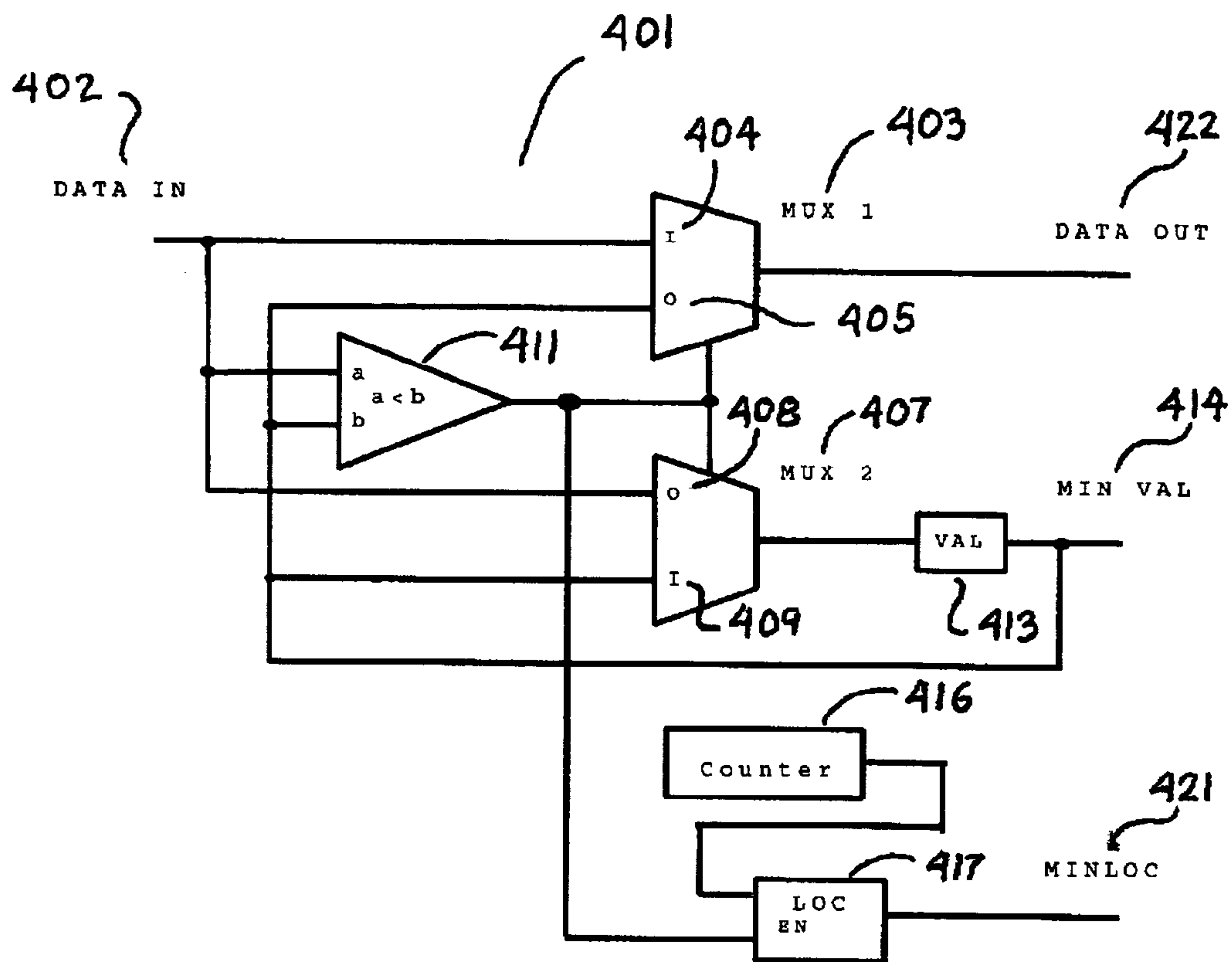


Fig. 11a

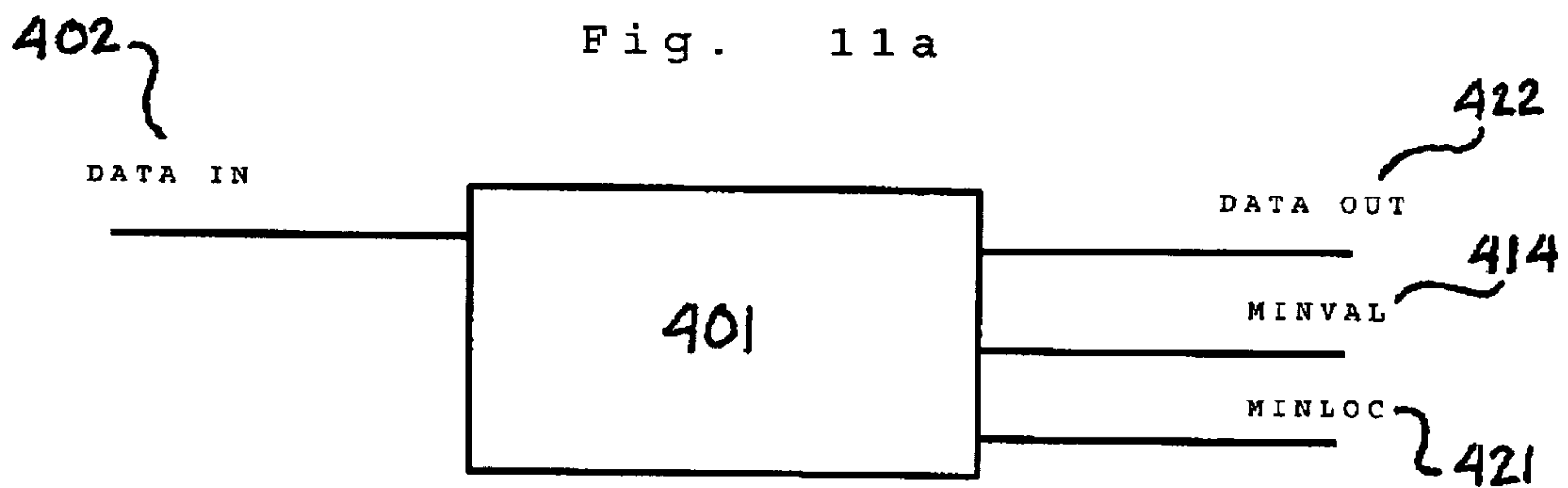


Fig. 11b
The minimum function

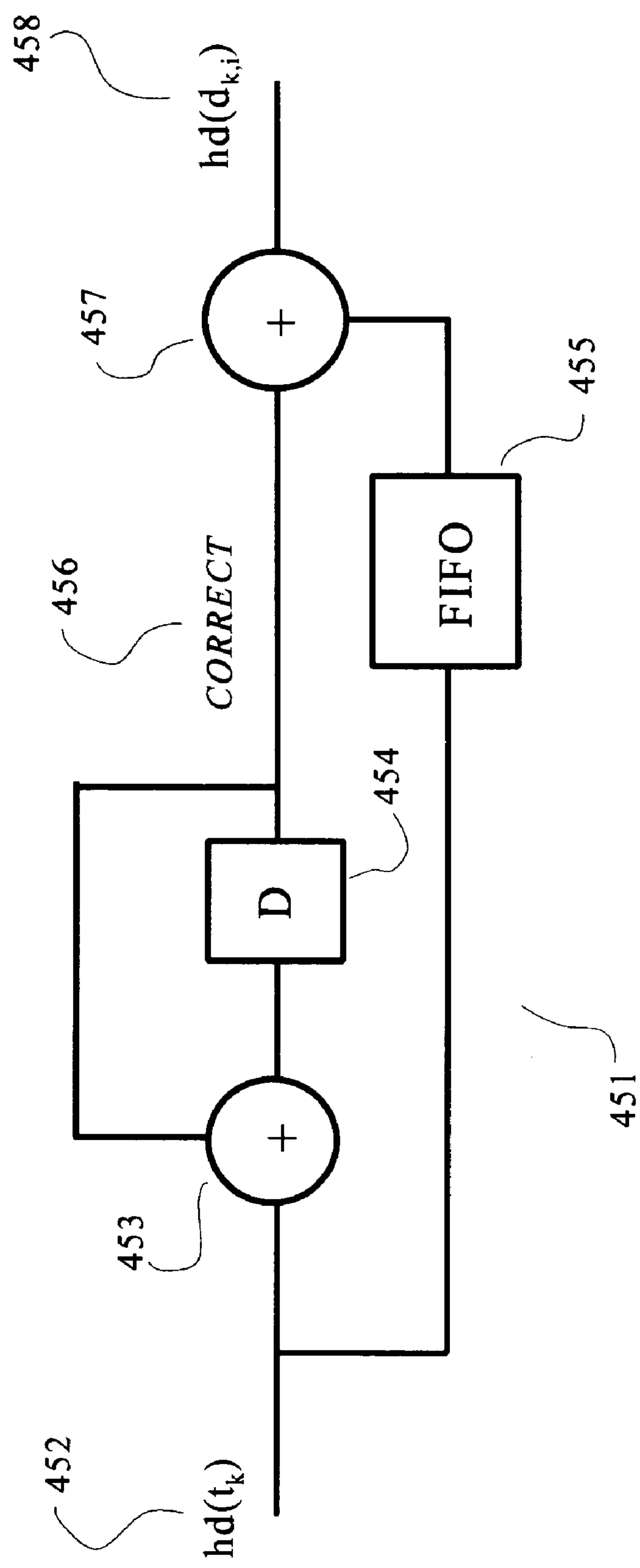


Fig. 12a

Sign bit data path of the SISO

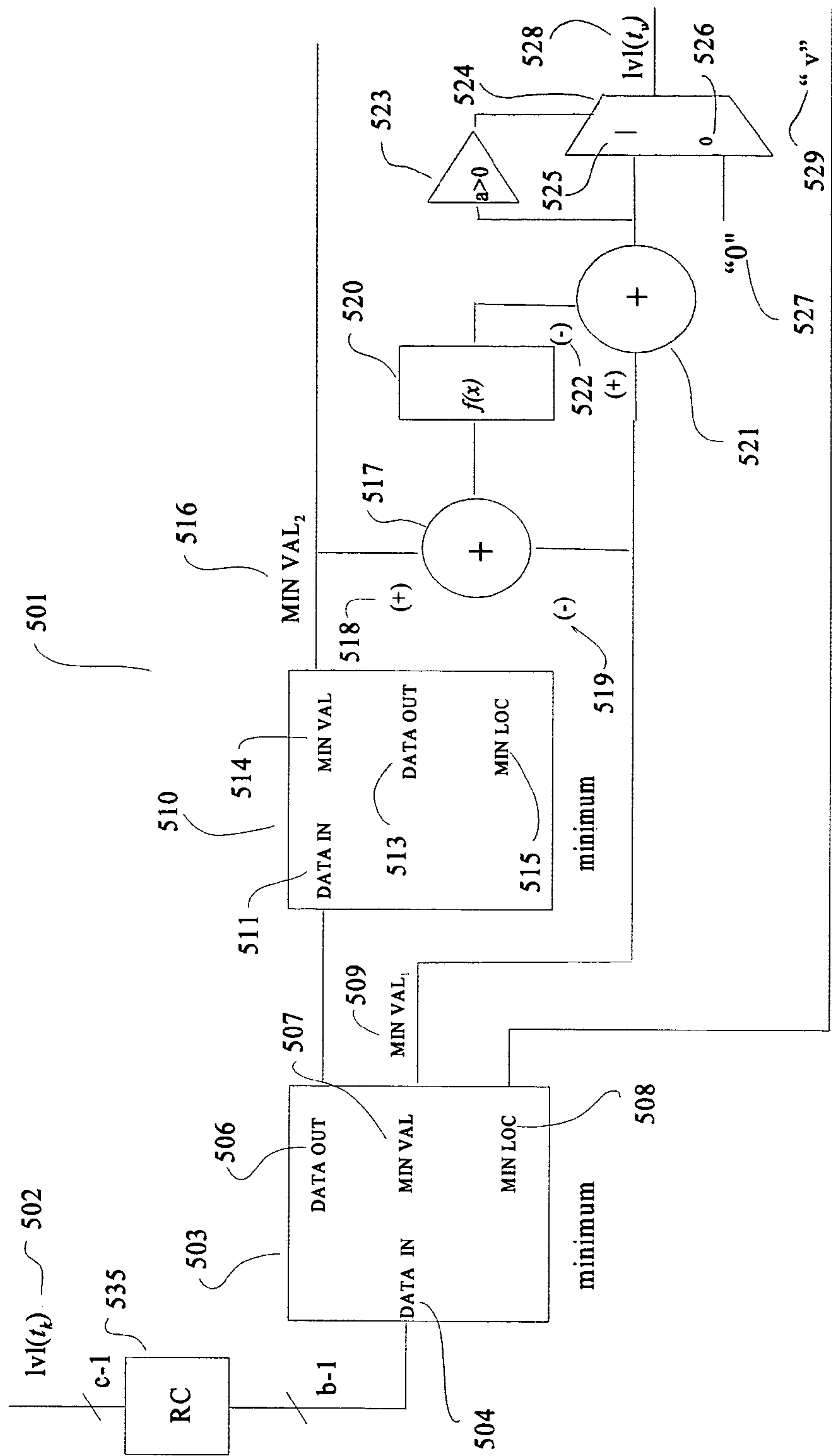


Fig. 12 b
Magnitude field data path of the SISO

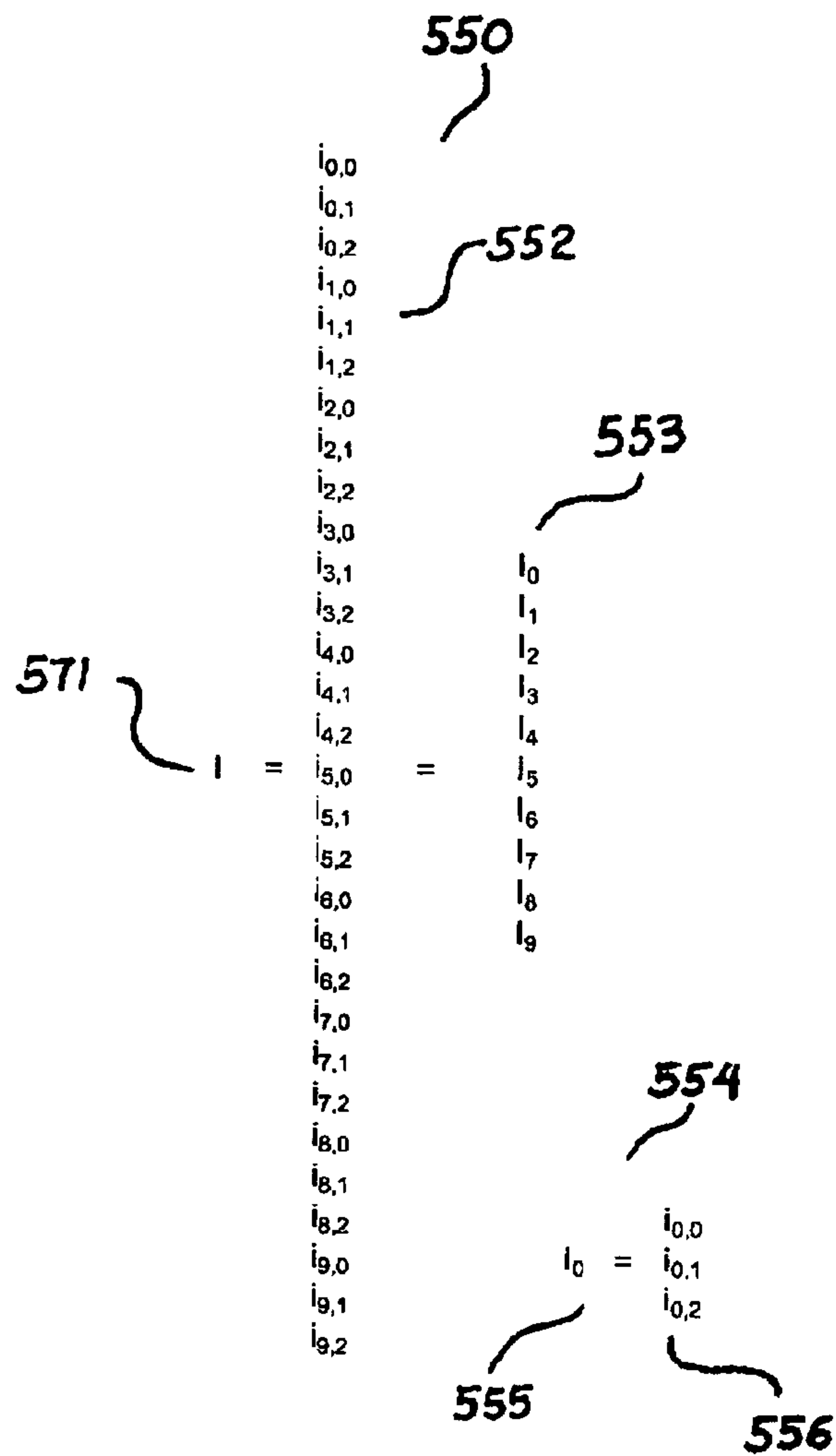


Figure 13

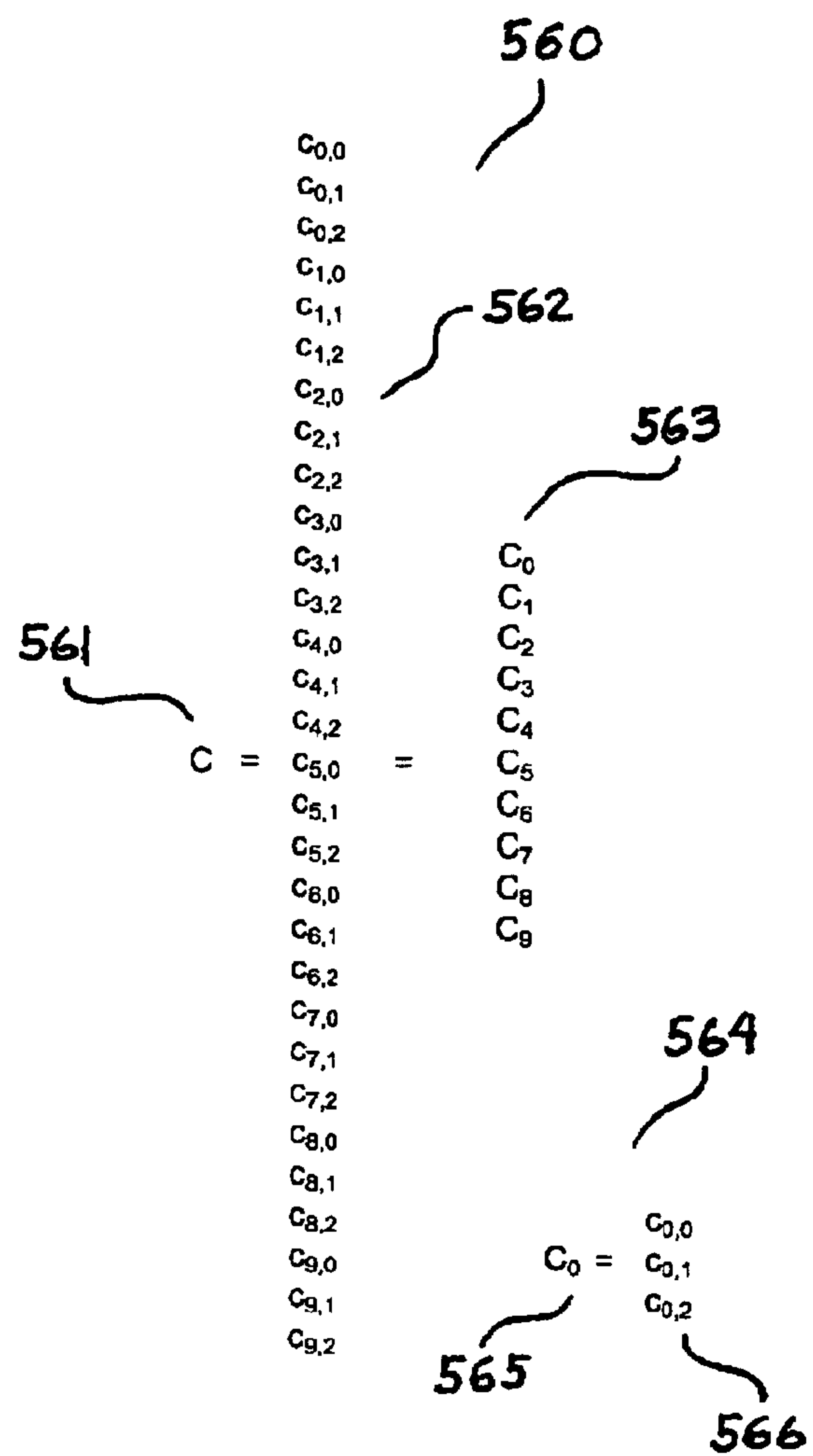


Figure 14

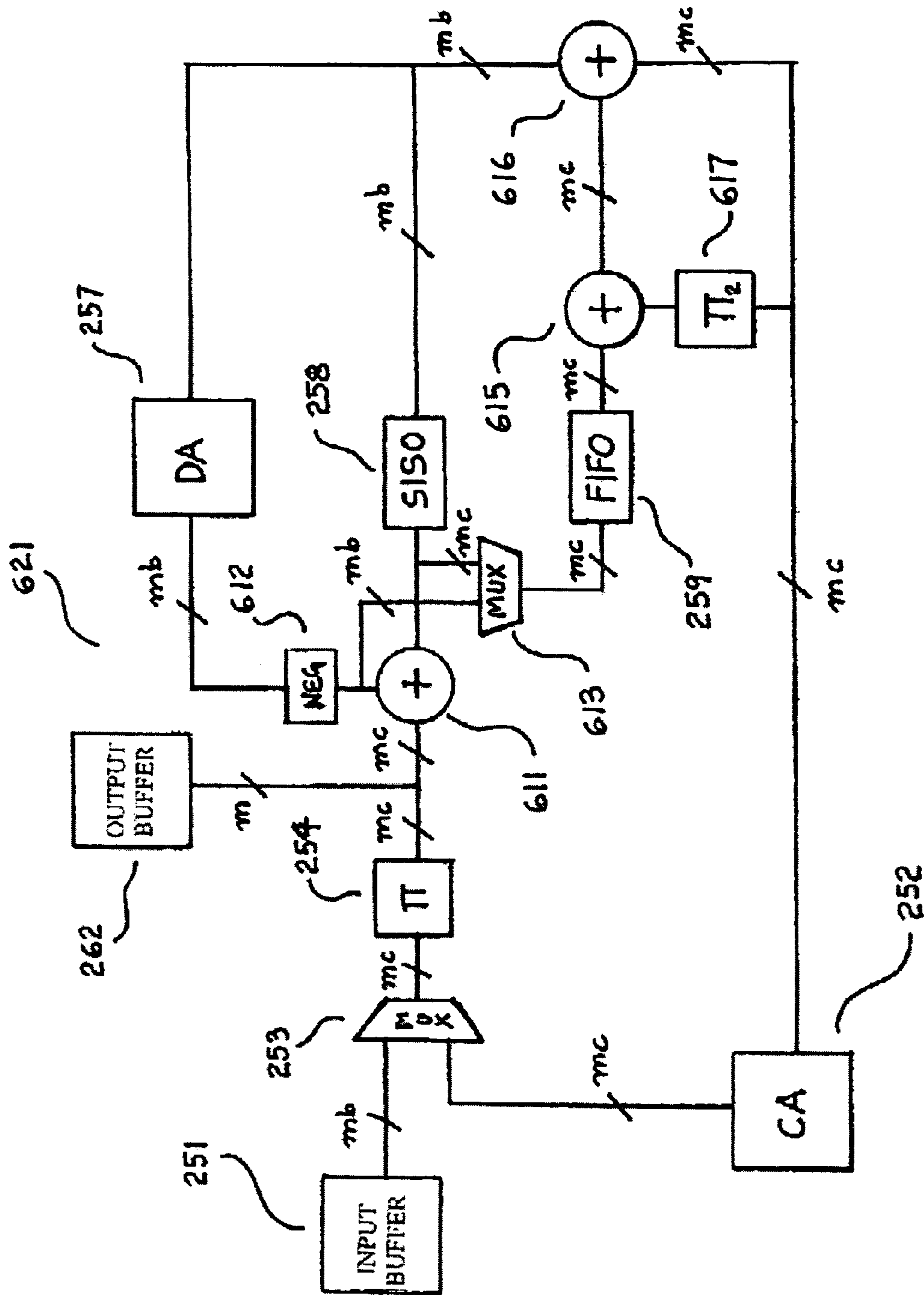


Fig. 15
m parallelism that handles multiplicities

SISO DECODER**CLAIM OF PRIORITY TO PROVISIONAL
APPLICATION (35 U.S.C. § 119(e))**

This application claims priority under 35 U.S.C. § 119(e) from provisional patent Application No. 60/568,939, filed May 7, 2004. The 60/568,939 Application is incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to error correction systems for computer data. More specifically, the invention relates to the use of parity check codes such as a low density parity check code ("LDPC").

BACKGROUND OF THE INVENTION

The transmission of binary computer data involves the introduction of errors, which must be detected and corrected, if possible. Although the difference between the two binary values, zero and one, seems clear, like the difference between black and white, in practice an electronic device may have difficulty distinguishing the difference. The difference between binary values may be detected as a voltage difference, but electronic noise in a circuit can interfere and render the difference less certain. This uncertainty must be dealt with. One option is to reject the data input and request retransmission. However, this is impossible with some fast flowing digital signals with substantial volume, such as digital TV, and is impractical in many situations. Accordingly, error correction systems have been developed to detect and correct errors. Communication systems often use forward error correction to correct errors induced by noise in the channel. In such systems, the error correction occurs at the receiver. One such system is parity check coding. One example of parity check coding is "low density parity check" coding ("LDPC")

Forward error correction consists of adding redundancy to data. Block codes, such as the LDPC codes, segment the data into blocks. These blocks have additional bits added according to a specified algorithm, to create a codeword. This codeword is transmitted to the receiver over the channel. The data that is transmitted is binary in nature, meaning that it is either a logical "1" or a logical "0". Noise is added by the channel, and the receiver detects each of the bits of the codeword and makes a best initial determination as to whether the bit is a logical 1 or 0. The receiver might also have the ability to assign a confidence in its guess. These guesses are called soft bits.

When a receiver gets a codeword, it is processed. The coding information added to original data is used to detect and correct errors in the received signal and thereby recover the original data. For received values with errors, the decoding system will attempt to recover or generate a best guess as to the original data.

As noted above, the receiver can reject data input containing errors. Retransmission may increase the reliability of the data being transmitted or stored, but such a system demands more transmission time or bandwidth or memory, and in some applications, such as digital TV signals, it may be impossible with current technology. Therefore, it is highly desirable to perfect error detection and correction of transmitted data.

LDPC systems use an iterative decoding process which is particularly suitable for long codewords. In general, LDPC codes offer greater coding gains than other, currently available codes. The object is to use parallel decoding in the

LDPC's iterative process to increase speed. In order to accomplish this, the inherent parallelism of an LDPC code must be found and exploited. There is also a need to reduce the amount of memory accesses and total memory required per iteration. To make the LDPC coding work as efficiently and quickly as possible, careful attention must be drawn to the storage of data and routing the data to the storage during the iterations.

U.S. Pat. No. 6,633,856 to Richardson et al. ("Richardson"), discloses two LDPC decoder architectures, a fast architecture and a slower architecture. In the slow architecture, a single iteration consists of two cycles. There is an edge memory consisting of one location for each edge in the Tanner Graph or, equivalently, there is one location for each 1 in the H matrix. There is also an input buffer which requires a memory location for each input variable, or equivalently, there is a memory location for each column of the H matrix. The two memories do not require the same resolution, the high resolution memory is the edge memory, and the low resolution memory is the input buffer. In the fast architecture, a single iteration consists of a single memory cycle. There are two edge memories and a single input buffer required.

SUMMARY OF THE INVENTION

The current invention involves a parallel SISO structure that allows the decoder to process multiple parity equations at the same time. There is a new SISO decoder which allows for the updating of the Log-likelihood-ratios in a single operation, as opposed to the two pass traditionally associated with the Tanner Graph's. In the decoder, there is a mapping structure that correctly aligns the stored estimates to the stored differences for presentation to the SISO's. There is also the ability to deal with multiple instances of the same data being processed at the same time. This structure manages the updates and the differences in such a manner that all calculations on a single piece of data that are processed in parallel are incorporated correctly in the new updated estimates.

The LDPC architecture of the present invention makes better use of memory and processing capacity during decoding. In the present invention, a single iteration consists of a single memory cycle. Two memories are disclosed. The first is a difference array which has a memory location for each of the ones in the H matrix, and the second is a current array which has a memory location for each of the columns in the H matrix. The current array may use high resolution memory, but the difference array requires only low resolution memory.

The LDPC architecture of the present invention requires the same number of memory cycles as the fast architecture of the Richardson architecture, but the present invention only requires the same number of memory locations as the slow architecture. Furthermore, the Richardson architectures require the larger memory to have higher resolution, while the present invention requires only the small memory as the higher resolution. The result is that, even with the same number of memory locations as the slow architecture of Richardson, the number of memory bits required by the present invention is less than required by even the slow architecture of Richardson.

Another significant difference between the present invention and the Richardson architectures is how permutations are handled. The Richardson architecture stores all the variable messages in their unpermuted form and the check messages in their permuted form. This requires a permutation block for each memory access. The architecture of the present invention represents the differences in their permuted form, and the variable nodes are stored in the same permutation as the last

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time they were accessed. They are permuted to the correct orientation each time they are used. The consequence is that only one permutation is required per iteration instead of the two required by the Richardson architecture. This is a significant savings, as the permuter is a fairly large function.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a representative parity check matrix (H-Matrix).
 FIG. 2 shows a signed magnitude data structure.
 FIG. 3a is a decoder architecture with no parallelism.
 FIG. 3b is a decoder architecture for expanded codes which allows for parallel processing of data.
 FIG. 4 shows all permutation transformations for 3 variables.
 FIG. 5a shows an expanded H-Matrix with permuted sets.
 FIG. 5b shows the H-Matrix of FIG. 5a without the zero blocks, for greater clarity.
 FIG. 6 is a matrix of permutations as an alternate representation for the H-matrix in FIG. 5a.
 FIG. 7 is a third alternate representation for the H-matrix showing the equations as sums of the input sets linked with their permutation.
 FIG. 8 is a table showing the result of two permutations.
 FIG. 9 shows the contents of the CA for each iteration of the expanded code.
 FIG. 10 shows the contents of the DA for the expanded code.
 FIG. 11a shows a circuit that finds the minimum value in a sequential list of values, and passes all the non-minimums through. It also gives the sequence number in the list of the minimum value.
 FIG. 11b shows the minimum function block.
 FIG. 12a shows the sign bit path of the SISO circuit.
 FIG. 12b shows the magnitude field path of the SISO circuit.
 FIG. 13 shows sets of inputs.
 FIG. 14 shows sets of Current estimates.
 FIG. 15 is a decoder architecture for expanded codes with the additional feedback path for handling multiplicities.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

1. The Coding Process

Communication systems often use forward error correction to correct errors induced by noise in a transmission channel. In such forward error correction systems, the detection and correction of errors occur at the receiver. Bits received through the channel are detected at the receiver as “soft” values. A soft value represents the “best guess” that the receiver can make for the value of the bit that was sent and the confidence in that guess. In essence, data is sent as a single bit, and received as a multi-bit sample. During transmission, a single bit of data may pick up noise, so that it is necessary to use more than a single bit to identify the sampled data. For example, in a binary system, if a “1” is coded as 5 volts and a “0” as 0 volts, then each can be represented with a single bit. If a value of 4.2 volts is received, then this is close to representing a “1”, but the receiver will use multiple bits to represent how close to the 5 volts the sampled data resides.

A typical format for the received data is signed magnitude, where the first bit is a sign bit representing the hard decision data, and the remainder of the bits represent the confidence in the hard decision bit. A “hard decision” is a single bit. In the example set out immediately above, the receiver reads 4.2 volts, but could output a “1” as a hard decision, which would indicate 5 volts. This is shown in FIG. 2 with the <hd> field 233 being a single bit hard decision, and the <lvl> field 234

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being a multi-bit confidence level. The signed magnitude is positive if the <hd> bit 232 equals one, and is negative if the <hd> bit 235 equals zero. An example of the signed magnitude format may be illustrated as follows:

	sign <hd>	magnitude <lvl>	
e.g.	1	00111	(representing a positive 7)
e.g.	0	00111	(representing a negative 7)

A type of forward error correction is low density parity check codes (LDPC). Low Density Parity Check codes are codes that have a “sparse” H-Matrix. A sparse H-Matrix is one in which there are many more zeroes than ones in the H-Matrix. For illustration here, a representative (non-sparse) H-Matrix 201 is shown in FIG. 1. The associated input vector “I” 209, representing the inputs i_k 210, and the current estimate of the sent data, vector “C” 220, are shown in FIG. 1 as well. Each row 202 of the matrix represents a parity equation. Each row is identified as the “ith” row, row_i 202. In FIG. 1, there are five rows 202, row 0 through row 4, in the exemplary H-Matrix. The number of inputs i 210 is equal to the number of columns in the H-matrix. In FIG. 1, there are ten columns in the exemplary H-Matrix; so, there are ten inputs i_k 210, i_0 through i_9 . The elements of the H-matrix are referred to as $H_{i,k}$ 200, which is the element in row i and column k .

In practice, an H-matrix will be much larger than the exemplary matrix of FIG. 1, and will have many hundreds if not thousands of data bits. By way of background information, an LDPC code is defined as “regular” if the H-matrix has the same number of 1’s in each column and the same number of 1’s in each row, and is “irregular” if it does not have the same number of ones in either the rows, the columns, or both. LDPC decoders work on “soft” channel data and are iterative in nature.

2. The SISO

As noted above, inputs are received in a signed magnitude representation. The inputs are stored in an input buffer 251 in FIG. 3a.

In its basic operation, the “Soft-In-Soft-Out” (“SISO”) function of an LDPC decoder evaluates each of the parity equations row_i 202, represented by the rows 202 of the H-Matrix 201 using the current estimates C 220, and if the parity equation is satisfied, will increase the confidence of the current estimates c_k 221 for those current estimates c_k 221 related to row_i 202. If the parity equation row_i 202 is not satisfied, the confidence of each current estimate c_k 221 related to row_i 202 will be decreased. It is possible to decrease the confidence to the point that a current estimate’s hard decision bit is actually flipped, producing a correction of erroneous data.

The parity equations that the SISO evaluates are determined by the multiplication of the H-Matrix 201 by the input vector I 210 and the multiplication of the H-Matrix 201 by the current estimate vector C 220. This multiplication yields the parity equations

$$i_0+1_1+i_3+i_5+i_9$$

$$i_1+1_2+i_4+i_5+i_6$$

$$i_0+1_2+i_3+i_6+i_7$$

$$i_0+1_1+i_4+i_7+i_8$$

$$i_2+1_3+i_4+i_8+i_9$$

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for the inputs and the parity equations

$$c_0 + c_1 + c_3 + c_5 + c_9$$

$$c_1 + c_2 + c_4 + c_5 + c_6$$

$$c_0 + c_2 + c_3 + c_6 + c_7$$

$$c_0 + c_1 + c_4 + c_7 + c_8$$

$$c_2 + c_3 + c_4 + c_8 + c_9$$

for the current estimates.

For each evaluation of a parity equation, the SISO outputs a difference for each of the inputs. This value is the difference between the input to the SISO and the estimate that this particular equation provides for that data. Referring to FIG. 3a, this difference is stored in the Difference Array ("DA") memory 257, as it is needed in subsequent evaluations of the same equation. The difference is also sent to an adder 260, where it is added to the data stored in the FIFO 259. This data is stored in the Current Array, "CA". Let c_k represent the current best estimate of the k th input to the decoder, and let $d_{i,k}$ represent the change in confidence to c_k according to parity equation i . The estimate c_k is stored in CA 252 and estimate $d_{i,k}$ is stored in DA 257.

The SISO 258 takes as inputs all the inputs identified by a row in the H-Matrix. As an example, for row 0 of the matrix in FIG. 1, inputs 221 c_0, c_1, c_3, c_5 , and c_9 are selected. The SISO 258 outputs a difference for each of the inputs; these are designated as $d_{0,0}, d_{0,1}, d_{0,3}, d_{0,5}$ and $d_{0,9}$ respectively. These are both stored into the DA memory 257, and added 260 to the original SISO inputs. The outputs of this adding operation are then stored back into the CA 252, replacing the values that were used in the equation.

After one complete iteration cycle, each of the parity equations, row 0 through row 4, will have been evaluated once, and the contents of the CA will be as follows:

$$c_0' = c_0 + d_{0,0} + d_{0,2} + d_{0,3}$$

$$c_1' = c_1 + d_{1,0} + d_{1,2} + d_{1,3}$$

$$c_2' = c_2 + d_{2,1} + d_{2,2} + d_{2,4}$$

$$c_3' = c_3 + d_{3,0} + d_{3,2} + d_{3,4}$$

$$c_4' = c_4 + d_{4,1} + d_{4,3} + d_{4,4}$$

$$c_5' = c_5 + d_{5,0} + d_{5,2}$$

$$c_6' = c_6 + d_{6,1} + d_{6,2}$$

$$c_7' = c_7 + d_{7,2} + d_{7,3}$$

$$c_8' = c_8 + d_{8,3} + d_{8,4}$$

$$c_9' = c_9 + d_{9,4} + d_{9,0}$$

The result c_k' is the new value for c_k which is stored back in the CA 252 after the iteration. The old value of c_k is overwritten by the new value.

The CA 252 will contain n signed magnitude values and the DA 257 contains as many signed magnitude values as there are 1's in the H-Matrix 201. In the above example, the DA 257 will have 25 entries, and the CA 252 will have 10.

a. SISO inputs/outputs

The data structure for c_k and $d_{i,k}$ is shown in FIG. 2. The bit <hd> 233 is the hard decision value, and <lvl> 235 is a multi-bit confidence value where the higher the value, the higher the confidence. The function $hd(c_k)$ returns the hard decision value of c_k , i.e., a "1" or a "0", and the function $lvl(c_k)$ returns the confidence value of c_k .

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Sticky adder 256 is placed ahead of the SISO 258. The sticky add function is defined as follows:

$$A \oplus B = A + B \text{ if } A + B < \text{MaxVal}$$

$$A \oplus B = \text{MaxVal} \text{ if } A + B \geq \text{MaxVal}$$

$$\text{MaxVal} \oplus B = \text{MaxVal} \text{ for all } B$$

Where A and B are variables and MaxVal is the maximum value that can be handled. For example, if X and Y are 6 bit signed magnitude registers, then the lvl field is a 5 bit number and the hd field is a single bit. If X is a positive 20 and if Y is a positive 15, then the binary value of X is 110100 and the binary value of Y is 101111. Then, $lvl(X) \oplus lvl(Y) = 31$.

There is an input i_k 210 and a current estimate c_k 221 associated with each column of the H-Matrix, and there is a difference associated with each non-zero entry in the H-Matrix; that is with every "1" entry. For example, when working on row 1 of the H-Matrix 201 in FIG. 1, the non-zero k 's are $\{1, 2, 4, 5, 6\}$. Each row of the H matrix represents one parity equation. When evaluating the equation represented by row i 202, the SISO takes as input t_k where

$$t_k = c_k \oplus (-d_{i,k}) \text{ for all } k \text{ where } H_{i,k} = 1$$

The value t_k is the output of adder 256 in FIG. 3a. It has the data structure that is shown in FIG. 2. From the adder 256, t_k is presented to the SISO 258, as well as stored in the FIFO 259.

The purpose of the SISO is to generate the differences. The differences are the differences between each input and current estimate as identified by the particular row equation being worked. The differences are defined by the following sets of equations:

$$\text{CORRECT} = \sum_k hd(t_k) \text{ where addition is over } GF(2)$$

$$\text{MinVal}_1 = \min(lvl(t_k)) \text{ for all } k$$

$$v = k : lvl(t_k) = \text{MinVal}_1$$

$$\text{MinVal}_2 = \min(lvl(t_k)) \text{ for all } k \neq v$$

$$hd(d_{i,k}) = hd(t_k) + \text{CORRECT} \text{ where addition is over } GF(2)$$

$$lvl(d_{i,v}) = \text{MinVal}_2$$

$$lvl(d_{i,k}) = \max(0, \text{MinVal}_1 - f(\text{MinVal}_2 - \text{MinVal}_1)) \text{ for } k \neq v$$

with the function $f(\text{MinVal}_2 - \text{MinVal}_1)$ is defined such as:

$$f(x) = \begin{cases} 3 & x < 2 \\ f(\text{MinVal}_2 - \text{MinVal}_1) = 2 & \text{for } 2 \leq x < 4 \\ 1 & 4 \leq x < 8 \\ 0 & 8 \leq x \end{cases}$$

The output of the SISO is $d_{i,k}$. This value of $d_{i,k}$ replaces the value that was read from the DA. The value of c_k that was read from CA is replaced with $t_k \oplus d_{i,k}$ for all k .

b. The minimum function

FIGS. 11a and 11b are block diagrams showing the minimum function of the present invention. FIG. 11b shows the input and outputs of the minimum function block 401, and

FIG. 11a shows the details of the minimum function block 401. The minimum function finds the minimum number in a list of numbers. The minimum number will be presented on the output MinVal 414 of the minimum function block 401. The sequence number of the minimum number will be presented on the output MinLoc 421. All other numbers in the sequence are passed through the minimum function block unchanged 422.

The minimum function block is initialized by having the counter 416 set to zero and the Val register 413 set to the maximum possible value with a preset which initializes the Val register 413 to all ones. The numbers are input on the Data_in line 402. This value is presented to the “a” input of the comparator 411. The “b” input of the comparator 411 is the current minimum value. After initialization, this is the maximum possible number. If “a” is less than “b”, then Mux 1 403 passes the Val register value to the output Data_out 422. Mux 2 407 passes the Data_in input 402 to the input of the Val register 413, where it is saved. If “a” is not less than “b”, then Mux 1 403 passes Data_in to the output Data_out 422. Mux 2 407 passes the contents of the Val register back to the Val register 413, in effect, leaving it the same.

As noted above, the counter 416 is initially set to zero. Every time new input is brought in, the counter is incremented. If Data_in 402 is less than the value stored in the Val register 413, the value of the counter 416 is latched into the Loc register 417. This corresponds to a new minimum value being stored in the Val register 413.

Once a sequence of numbers have passed through the minimum function block, the output MinVal 414 has the minimum value and the output MinLoc 421 has the location in the sequence of the minimum value.

By way of example, if the sequence {14, 16, 10, 10} were passed through the circuit, the following would occur. The counter 416 is initialized to zero and the Val register 413 is initialized to a maximum value. The number 14 is input. 14 is less than a maximum value, so 14 gets placed in the Val register 413, the number 0 is placed in Loc 417, and the maximum value is passed to the output Data_out 422 and the counter 416 is incremented to 1. Then the number 16 is input. 16 is larger than the 14 that is in Val 413 register, so the Val register 413 maintains its value of 14, the register Loc 417 maintains its value of 0, 16 is passed to the output Data_out 422 and the counter 416 is incremented to 2. Then the number 10 is input. 10 is less than the 14 that is in Val register 413, so the Val register 413 is changed to 10, the number 2 is placed in Loc 417, 14 is passed the out Data_out 422 and the counter 416 is incremented to 3. Then the second number 10 is input. The second 10 is not less than the first 10, so the 10 that is in Val register 413 stays the same, the value of Loc 417 does not change, the second 10 is passed out Data_out 422 and the counter 416 is incremented to 4. As this is the end of the sequence, the MinVal output 414 is 10 and the MinLoc output 421 is 2.

C. Details of the SISO

The SISO is shown in FIG. 12a and 12b. It takes as its input the string of t_k 's. Each of the values is a signed magnitude number. These inputs are $hd(t_k)$ 452 in FIG. 12a, which is the sign bit, and $lvl(t_k)$ 502 in FIG. 12b, which is the magnitude portion of the number. The SISO deals with these portions separately, and they are recombined at the output. As such, they will be dealt with as separate data paths. FIG. 12a is the sign bit data path and FIG. 12b is the magnitude field data path.

First consider the sign bit data path in FIG. 12a. The flip flop 454 in FIG. 12a is initialized to 0. As each sign bit is input, it is exclusive-or'd with contents of the flip flop 454 and

the result is placed in the flip flop 454. After all the $hd(t_k)$'s 452 are input, the flip flop 454 contains the exclusive-or of all the sign bits. This is the signal named “CORRECT” 456. Each of the sign bits are also stored in a FIFO 455. Once all the $hd(t_k)$'s 452 have been input, the CORRECT bit 456 is fixed. This bit is exclusive-or'd with each of the sign bits that has been stored in the FIFO. These become the sign bits for the new $hd(d_{i,k})$'s 458 that are output from the FIFO 455. Thus, if the parity of the inputs is even, each of the new $hd(d_{i,k})$'s 458 will have the same sign as the respective $hd(t_k)$ 452. If the parity of the inputs is odd, then each of the new $hd(d_{i,k})$'s 458 will have the opposite sign as the respective $hd(t_k)$ 452.

The magnitude or confidence data path is shown in FIG. 12b. The confidence values of the $lvl(t_k)$'s 502 are brought into the block c-1 bits wide. They are converted to b-1 bits wide in the resolution converter block RC 535. If the most significant c-b bits are zero, then the least significant b-1 bits are passed through unchanged. If any of the most significant bits are 1, then the least significant b-1 bits are set to 1. In effect, if the input $lvl(t_k)$ 502 can be represented with b-1 bits, then it is so represented, otherwise, it is set to the maximum value that can be represented with b-1 bits. This output is input to a first minimum block 503 where both the minimum value, MinVal₁ 507, and its location, “v” 529, are found and stored for outputs. The Data_out 506 from the first minimum block 503 is the input 511 to the second minimum block 510, where the second lowest value of the string of confidences is found.

The first sum block 517 takes MinVal₁ 509 and MinVal₂ 516 as inputs, with MinVal₁ 509 as a negative input 519. The output of the first sum block 517 is input to the f(x) block 520. The f(x) block 520 has A function listed SUCH as

$$f(x) = \begin{cases} 3 & x < 2 \\ 2 & \text{for } 2 \leq x < 4 \\ 1 & 4 \leq x < 8 \\ 0 & 8 \leq x \end{cases}$$

This output is input to the second sum block 521 as a negative input 522. The other input is MinVal₁ 509. The output of this second sum block 521 is input to a comparator 523, as well as input to a Mux 524. The Mux 524 has a second input which is a zero value 527. The comparator 523 tests to see if the input is greater than 0. The output of the comparator 523 is the select input of the Mux 524. If the comparator 523 tests true, then the output of the second sum block 521 is passed to the output as the $lvl(t_v)$ output 528. If the comparator 523 is false, then the zero input 527 is passed to the output as the $lvl(t_v)$ output 528. Finally, MinVal₂ 516 is passed to the output as the MIN($lvl(t_k)$) output for k equal to v.

In summary, referring to FIG. 3a, when a row is being evaluated, all the differences associated with that row are removed from the current best estimates prior to evaluating the new differences. This result is also stored in the FIFO 259. The new differences are stored in DA 257 and also added 260 with the output of the FIFO 259 prior to being stored back into CA 252. For any row operation, this amounts to a replacement of the current estimate or the current difference with the new estimate or the new difference.

FIG. 3a shows a circuit that performs this function.

3. Expanded Code

The H-Matrix in FIG. 1 can be viewed as a mother code. Each of the 1's in that H-Matrix can be replaced with an $m \times m$ permutation matrix and each zero can be replaced with an $m \times m$ zero matrix. This will allow the decoder to operate on sets of inputs m at a time. The decoder will access differences m at a time, current estimates will be accessed m at a time, and there will be m SISO's. The parameter m is known as the set size of the decoder.

As an example, let $m=3$. In such a case, there are 6 possible permutations, any of which can be used. These permutations are shown in FIG. 4, which also illustrates the effect of a permutation. For example, with Permutation P_2 , the input of abc is permuted to acb. An example of an expanded H-matrix with each of the permutations included is shown in FIG. 5a. In FIG. 5b, the 3×3 zeros have been blanked out for clarity.

FIG. 8 shows the effect of a permutation on a set that is already permuted. The columns of "A" permutations 361 are mapped against the rows of "B" permutations 362. Thus, for example, if A permutation 361 is permutation P_2 , the second column, is followed by permutation P_3 , the B permutation 362 in the third row, that has the same effect as permutation P_4 . As another example, permutation P_3 followed by permutation P_2 has the same effect of permutation P_5 . If a block is sitting in permutation P_x and needs to be mapped to permutation P_y , the table in FIG. 8 can be used to determine the necessary permutation to apply. For example, if a block is in permutation P_4 , and needs to end up in permutation P_2 , looking at the table, it will be seen that permutation P_3 will give the desired result. As a second example, if a block is in permutation P_5 and needs to end up in permutation P_1 , then permutation P_4 will accomplish that result.

Each of the equations, the differences, the inputs and current estimates will be grouped in sets of m . Looking at the Matrix in FIG. 5b, row $i_{j,l}$ is the j th row in set Row $_i$. Input $i_{k,l}$ is the l th input in set I_k . This grouping is shown in FIG. 13, and is analogous to the current estimate, which is shown in FIG. 14. Thus, referring to FIG. 13, input $i_{k,l}$ 552 is the l th input in set I_k 553; referring to FIG. 14, input $c_{k,l}$ 560 is the l th current estimate in set C_k 563. The differences are also grouped into sets of m . The individual differences have been referred to as $d_{i,k}$. The set of differences associated with Row $_i$ and C_k are referred to as $D_{i,k}$. There are m differences, again associated with the ones in the H-Matrix.

Another exemplary representation for the H-matrix is shown in FIG. 6. Each zero represents a 3×3 zero matrix, e.g. 331, and each number represents the 3×3 permutation matrix, e.g. 332, from FIG. 4. Thus, the number in FIG. 6 refers to one of the six possible permutations, P_1 through P_6 , identified in FIG. 4. For example, in the first row and first column of FIG. 6, permutation 2, or P_2 , is identified. In permutation 2, the input "abc" is permuted to the output "acb". The 3×3 matrix for P_2 is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Finally, a third representation is listed in FIG. 7. In this case, each row 342 represents three equations. Each equation has 5 terms 341, where the term (k,m) 343 indicates that it is input set k with permutation m .

The purpose of the decoder architecture is to allow parallel solution of equations and allowing for a wider memory struc-

ture and reads that are more than one input wide. The decoder shown in FIG. 3b will accomplish this. The data paths are each m -inputs wide. The exemplary parallel architecture shown in FIG. 3b illustrates parallelism by showing three parallel paths. These three parallel paths lead to three parallel SISO's, 258₁, 258₂, and 258₃. However, the use of three parallel paths is merely illustrative, and the invention may, and usually will, employ many more paths. Thus, the use of three paths in this disclosure is not limiting, but exemplary.

The DA 257 is the memory that holds all the $D_{i,k}$'s. They are in groups of m , and stored in the "proper" order, where the "proper" order means the permutation indicated by that permutation matrix of FIG. 6. Anytime a current estimate is brought to an adder 256₁₋₃ to have the differences subtracted, the current estimate is permuted to the same permutation as $D_{i,k}$. As an example, the contents of $D_{0,0}$ for the matrix in FIG. 5b are stored in the order {1,3,2}, as indicated by the permutation P_2 .

An example working through two complete iterations for the code defined by the H-Matrix in FIG. 5b and the decoder of FIG. 3b is given below. The first time an input is used, a mux 253₁₋₃ selects the input 251; for subsequent uses of that "input", a mux 253₁₋₃ selects the CA 252. Each of the inputs 251 arrives in permutation order P_1 . Each of the C_i 's also need to be permuted to the proper order. An I_0 is permuted by P_2 and presented to a SISO 258₁₋₃ as well as stored in a FIFO 259₁₋₃. No difference is removed, as this is the initial pass through the decoder and the differences are all zero. The next inputs are I_1 permuted by P_3 , I_3 permuted by P_5 , I_5 permuted by P_1 , and I_9 permuted by P_4 .

Referring to FIG. 5b, and looking at just the equation represented by row_{0,0}, it requires inputs $i_{0,0}$, $i_{1,1}$, $i_{3,1}$, $i_{5,0}$ and $i_{9,2}$. By the same token, the equation represented by row_{0,1} requires inputs $i_{0,2}$, $i_{1,0}$, $i_{3,2}$, $i_{5,1}$ and $i_{9,0}$ and the equation represented by row_{0,2} requires inputs $i_{0,1}$, $i_{1,2}$, $i_{3,0}$, $i_{5,2}$ and $i_{9,1}$. Each of these inputs is presented to a proper SISO 258₁₋₃, by the permutations referenced in the paragraph above.

The differences calculated by a SISO 258₁₋₃ are stored in the DA 257 as $D_{0,0}$, $D_{0,1}$, $D_{0,3}$, $D_{0,5}$, and $D_{0,9}$. These differences are also added 260₁₋₃ to the inputs stored in the FIFO 259₁₋₃ and stored back in the CA 252. Note that the inputs are now stored back in the original location, but in a permuted form.

This exemplary architecture allows three SISO's, 258₁, 258₂, and 258₃, to operate in parallel. The inputs are read three at a time.

As the equations for the remaining ROW $_i$'s are evaluated, there is always a choice in taking the input from C_k or I_k . If I_k has been used, then select C_k . If I_k has not been used, then select I_k . This can be seen by examining ROW $_1$. I_1 has been used, so C_1 is selected by a mux 253₁₋₃. C_1 needs to be permuted to P_6 . However, it is already permuted by P_3 . Permutation P_4 accomplishes this. Therefore, C_1 is permuted by P_4 . I_2 has not been used, so it is selected by the mux 253₁₋₃. I_2 is permuted by P_4 . By the same token, I_4 has not been used, so I_4 is permuted by P_3 , I_5 has already been used, so C_5 is selected by the mux 253₁₋₃ and permuted by P_1 . I_6 has not been used, so I_6 is selected by the mux 253₁₋₃ and permuted by P_6 . Note that inputs I_2 , I_4 and I_6 were in their initial states, as they had not yet been permuted. With respect to the three SISO's, SISO₀ 258₁ gets $i_{1,2}$, $c_{2,2}$, $i_{4,1}$, $c_{5,0}$ and $i_{6,0}$; SISO₁ 258₂ gets $i_{1,1}$, $c_{2,0}$, $i_{4,0}$, $c_{5,1}$ and $i_{6,1}$; SISO₂ 258₃ gets $i_{1,0}$, $c_{2,1}$, $i_{4,2}$, $c_{5,2}$ and $i_{6,2}$.

The differences $d_{1,1}$, $d_{1,2}$, $d_{1,4}$, $d_{1,5}$ and $d_{1,6}$ are all initially zero. The new differences are stored in the DA 257. The differences are also added 260₁₋₆ into the output of a FIFO 259₁₋₃, which are then stored in the CA 252 as C_1 , C_2 , C_4 , C_5

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and C_6 respectively. This continues for ROW_2 , ROW_3 , and ROW_4 , at which point each of the equations has been solved once. At this point, the DA 257 is filled with non-zero values.

In general, the proper permutation to perform on any C_j can be determined by looking at the H-Matrix of FIG. 5b. Each C_j is stored in the permutation required by the equation that used the C_j previously. As an example, to determine the proper permutation for C_4 in ROW_3 equation, it can be observed the prior use of C_4 is in Row_1 . The permutation that C_4 is stored in is P_3 . Permutation P_6 is required, and P_4 is the permutation that accomplishes this. A second example would be C_4 in ROW_2 . Except for the very first iteration, which has already been dealt with, C_4 will be stored in permutation P_4 , which is seen by looking at C_4 in ROW_4 . Permutation P_6 accomplishes the required permutation.

This architecture keeps the differences $D_{i,k}$ in the permutation that is seen in the H-Matrix of FIG. 5b. The inputs are originally in permutation P_1 , which is no permutation. However, the permutation changes during the iterations. This can be seen in FIGS. 9 and 10. FIG. 9 identifies the permutation that each of the CA blocks is stored in after each of the 5 iteration steps. FIG. 10 shows the permutations for the DA memory. The required permutation at each step can be determined from these two figures. These document the permutation that the C_k is stored in, and the permutation that is needed. FIG. 8, as seen earlier, can be used to determine the proper P_x that is needed. For example, referring to FIG. 9, when working on ROW_3 equation, C_0 is stored in the P_4 permutation, which can be seen from the ROW_2 column. From FIG. 10 it may be seen that $D_{3,0}$ is stored in the P_5 permutation. Referring to FIG. 8, it can be seen that by applying permutation P_4 to C_0 , it will be in the proper P_5 permutation. When the new C_0 is calculated, it is stored in the P_5 permutation.

To get the required outputs, everything needs to be permuted back to P_1 . At the end of the last iteration, C_0 and C_2 are stored in permutation P_5 , C_1 and C_3 are stored in permutation P_2 , C_4 is stored in permutation P_4 and the rest are stored in permutation P_1 . C_0 and C_2 are permuted by P_4 , C_1 and C_3 are permuted by P_2 , C_4 is permuted by P_5 and the rest are permuted by P_1 . This gets all the outputs into the required P_1 permutation.

4. Multiplicity Architecture

One of the conditions that can occur in a code is when the same set of inputs is used more than once in the same set of equations. The input sets will occur with different permutations. An example would be to replace the second term in ROW_0 with (3, 4). The equation becomes

$$\text{Row } I = (0,2) + (3,4) + (3,5) + (5,1) + (9,4)$$

This requires I_3 to be used twice in the first iteration, followed by C_3 being used twice in subsequent iterations. The terms (3, 4) and (3, 5) are called "multiplicities" and occur when the same set of inputs are used in the same equations more than once. When this occurs, the input set will always occur with different permutations. There is a difference stored in the DA for each of these permutations. In the above example, the first difference has permutation P_4 , and the second difference has permutation P_5 . $D_{4,0,3}$ will represent the difference set $D_{0,3}$ with permutation P_4 and $D_{5,0,3}$ will represent the difference set $D_{0,3}$ with permutation P_5 . In general, $D_{y,i,k}$ will represent the set associated with the equation for ROW_i and the input set C_k in permutation P_y . Each of these is a separate set of differences, and will be stored separately in the DA 257. However, both differences require the same input, which is not permitted. In the first iteration, the input vector I 251 will be selected by the mux 253, and in subse-

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quent iterations the current estimates C will be stored in the CA 252 will be selected by a mux 253₁₋₃. The output of the mux 253 minus the first difference will be stored in a FIFO 259₁₋₃, as well as the output of the mux 253₁₋₃ minus the second difference. After doing multiple operations on the same C_k in the same iteration, the decoder in FIG. 3b is not capable of storing the proper value back in the CA 252. This is remedied by adding a feedback path, which is shown in FIG. 15. FIG. 15 shows an exemplary parallel system of width m . Thus, the system may use m parallel SISO's 258 to receive inputs from m paths. Similarly, the system may use m parallel mux's 253, m parallel adders 611, etc. In this way, FIG. 15 illustrates the parallelism of the present architecture in a manner distinct from the parallelism illustrated in FIG. 3b.

When processing ROW_0 for the first time, the first input is I_0 with a permutation of P_2 . The second input is I_3 with a permutation of P_4 . The third input is I_3 , but with a permutation of P_5 . The fourth and fifth inputs are I_5 and I_9 with permutations of P_1 and P_4 respectively. The inputs to a FIFO 259 are also different when dealing with repetitive sets in the same equation. The first time a set element is seen, the FIFO 259 receives $I_3 - D_{4,0,3}$. Recognize that $D_{4,0,3}$ is zero, as this is the first pass, and the differences are initialized originally to zero. The next input 251 to the FIFO 259 will be $(-D_{5,0,3})$. Again, $D_{5,0,3}$ is equal to zero. Also, $D_{5,0,3}$ is stored in a different location in the DA 257 than $D_{4,0,3}$, which allows for the retention of both values. When $I_3 - D_{4,0,3}$ is output from the FIFO 259, the other input to the adder 615 will be equal to zero. The output of adder 1 615 goes to adder 2 616. The new $D'_{4,0,3}$, which is output from the SISO 258, is added to the output of adder 1 615 using adder 2 616. This goes to the second permutation block 617, where it is permuted to P_5 . Referring to FIG. 8, it can be seen that this is accomplished with permutation P_4 . This is added 615 to the negative of the original difference $-D_{5,0,3}$ as it is output from the FIFO 259, and added by adder 616 to the new difference $D_{5,0,3}$. This result is stored into the CA 252 in location C_3 . The value that is stored is:

$$C'_3 = C_3 - D_{4,0,3} + D'_{4,0,3} + (-D_{5,0,3}) + D_{5,0,3}$$

This shows that both differences have been updated with the new values.

For subsequent iterations, C_3 will be stored in permutation P_2 . In that case, when ROW_0 is evaluated, C_3 will first be brought in with permutation P_3 , and second brought in with permutation P_6 to get the required permutations of P_4 and P_5 respectively.

In the multiplicity architecture shown in FIG. 15, the second permutation block 617 feeds a permuted version of the output of adder 2 616 back to be accumulated with the output of FIFO 259. In an alternative embodiment, the second permutation block 617 includes a delay. This delay provides the ability to pick more precisely the time of the feedback.

5. Processor Architecture

The architecture disclosed here arranges components as discrete components, but this is illustrative and not intended to be limiting. FIGS. 3a, 3b and 15 show the SISOs, FIFOs, DA, adders and CA, as separate blocks for illustrative purposes. This should not be construed as requiring a decoder with discrete components for these functions. For example, a decoder processor could integrate the SISO 258, FIFO 259 and second adder 260 of FIG. 3a into a single processor, or it could separate these functions into discrete components, and both arrangements would fall within the understanding of the invention disclosed and claimed here. Similarly, for example, the difference and new estimate functions of the SISO could

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be separated or integrated, and each arrangement would fall within the scope of the present invention.

The figures and description set forth here represent only some embodiments of the invention. After considering these, skilled persons will understand that there are many ways to make an SISO decoder according to the principles disclosed. The inventors contemplate that the use of alternative structures or arrangements which result in an SISO decoder according to the principles disclosed, will be within the scope of the invention.

The invention claimed is:

1. A SISO decoder for a parity check equation comprising:
 - a. a plurality of values indicating relative confidences for bits in a parity check equation, and
 - b. one or more functions mapping two or more minimum values from the plurality of values and generating magnitude updates for all relative confidences, wherein the magnitude mapping functions are further comprised of a first mapping not using a first lowest value when generating the magnitude for a bit location corresponding to the first lowest value, and
 - a second mapping using at least the first lowest value and a second lowest value when generating the update for any bit location not corresponding to the first lowest value.
2. The SISO decoder of claim 1, wherein the first mapping is proportional to the second lowest value, and the second mapping is proportional to a result of the first lowest value reduced by a magnitude reduction associated with the second lowest value and the first lowest value, and wherein the second mapping is further constrained to be non-negative.
3. The SISO decoder of claim 2, wherein the magnitude reduction of the second mapping is equal to a nonlinear differences function of the differences of the first and second lowest values.
4. The SISO decoder of claim 3, wherein the nonlinear differences function is comprised of the following definitions:
 - a. equal to 0 if the differences of the first and second lowest values is greater than 7,
 - b. equal to 1 if the differences of the first and second lowest values ranges from 4 to 7 inclusive,
 - c. equal to 2 if the differences of the first and second lowest values ranges from 2 to 3 inclusive, and
 - d. equal to 3 if the differences of the first and second lowest values is less than 2.
5. A decoder to evaluate parity equations, comprising:
 - a. a plurality of confidence values, the confidence values further comprising two or more minimum confidence values from the plurality of confidence values,

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- b. a first minimum function mapping a second lowest value from the plurality of confidence values and a location of a lowest value, and
- c. a second minimum function mapping at least the lowest value from the plurality of confidence values, wherein the second minimum function further maps confidence values for one or more locations not corresponding to the first lowest value.
6. The decoder of claim 5, wherein the second minimum function generates a nonnegative update proportional to a result of the lowest value reduced by a magnitude reduction associated with the second lowest value and the lowest value.
7. The decoder of claim 6, wherein the magnitude reduction of the second mapping function is a nonlinear function of the differences of the lowest value and the second lowest value.
8. The decoder of claim 5, wherein the first minimum function further comprises a circuit to receive the plurality of confidence values, said circuit comprising:
 - a. a counter to count the location of each confidence value,
 - b. a comparator to compare the values of the plurality of confidence values and find the lowest value, and
 - c. a register to store the lowest value.
9. A method for updating confidence values using two or more minimum confidence values from a string of confidence values, comprising the steps of:
 - a. receiving a plurality of confidence values,
 - b. first mapping the plurality of confidence values not using a first lowest value,
 - c. generating a magnitude update for a location corresponding to the first lowest value,
 - d. subsequently mapping the first lowest value and a next lowest value and generating the magnitude update for any location other than the first lowest value,
 - e. outputting the magnitude updates for at least the lowest and next lowest values.
10. The method of claim 9, wherein the first mapping is proportional to the next lowest value, and the subsequent mapping is proportional to a result of the first lowest value reduced by a magnitude reduction associated with the next lowest value and the first lowest value.
11. The method of claim 10, wherein the subsequent mapping is constrained to be non-negative.
12. The method of claim 10, wherein the magnitude reduction of the subsequent mapping is a nonlinear function of differences of the lowest value and the next lowest value.

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